

Lab Environment setup

7BUIS022W

University Of Westminster

1 Software, Tools and Files you need

Obtaining the machines

- For this module we will be using two different virtual machines. You can setup the lab environment either on your own device or using a lab machine.
- If you are using the machines in CLG.43 or CLG.45, you do not need to install VirtualBox or 7zip. They are already pre-installed. You can go to section [1.2](#)
- On your own machine, you first need to download and install the tools we need. Check section [1.1](#)
- **If you are having a problem, go to section [5](#). If this is still not working, please check with your lab instructor.**

1.1 Download & Install tools

1. Virtualisation software: VirtualBox

- Information about VirtualBox is available on this [link](#).
- You can download virtualbox from the [VirtualBox download page](#)
 - You need to make sure you download it for the right environment (Windows, Linux, macOS)

2. Compression software: 7-Zip

- You will also need to download 7-zip software to extract 7zip files.
- Information about 7-Zip is available on this [link](#).
- You can download 7-Zip from the [7-Zip download page](#)
 - You need to make sure you download it for the right environment (Windows, Linux, macOS)

1.2 Obtain the virtual machines

- For this module we will be using two different virtual machines.
- **On your own device you will need to download this environment once - see section ??**
- **On a university machines in CLG.43 and CLG.45 4 - You will need to make sure that you have the right VMs.**
- The two Virtual Machines we use are:
 - 1- **OWASP Broken Web Apps:** This is OAWSP VM, the vulnerable machine that contains all the different services (Apache, databases, web apps, etc..).

- 2- **kali Linux**: This is Kali Linux VM, the attacker or the penetration tester will be using this. It contains several pre-installed tools that hackers and penetration testers use.
- Each virtual machine will come compressed using 7zip files.
 - 1- **OWASP_Broken_Web_Apps_1.2.7z**
 - 2- **kali-linux-2023.2A-UoW-VBox-amd64.7z**
 - To download them, open the browser and please choose from below:
 1. **If you are using a university machine in the lab**
 - Check first that the 7zip files are NOT already in C:\VirtualMachines
 - It is more likely that someone else did this lab before you and the files exist.
 - If you find the files and some folders. It is safe to delete the folders but please do **NOT** delete the 7zip files.
 - **For CLG.43**
 - * <http://192.168.143.203/download>
 - **For CLG.45**
 - * <http://192.168.145.206/download>
 2. **If you are working from home:**
 - **Access from the ECS download page**
 - * <https://download.ecs.westminster.ac.uk/VirtualMachines/>
 3. **If you are your own machine and**
 - **You are connected to Eduroam:**
 - <http://10.20.144.78/download/VMs/SecurityVMs/>

2 Setting up the machines

- (1) After downloading (or checking) that VM files,
- (2) Browse to the VirtualMachines folder C:\VirtualMachines
- (3) You will have two files as shown in Fig.1. These are the two VMs files in a compressed form (.7z) to save space.

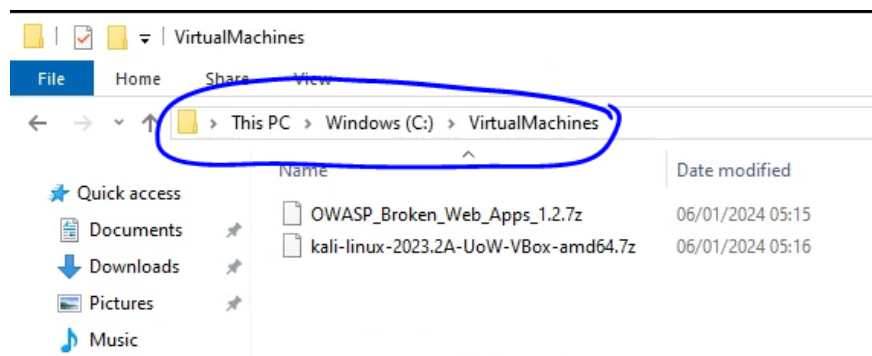


Figure 1: Virtual Machines folder

- (4) Right click on **kali-linux-2023.2A-UoW-VBox-amd64.7z** and select **7-Zip**, then select **Extract Here** as shown in fig.2

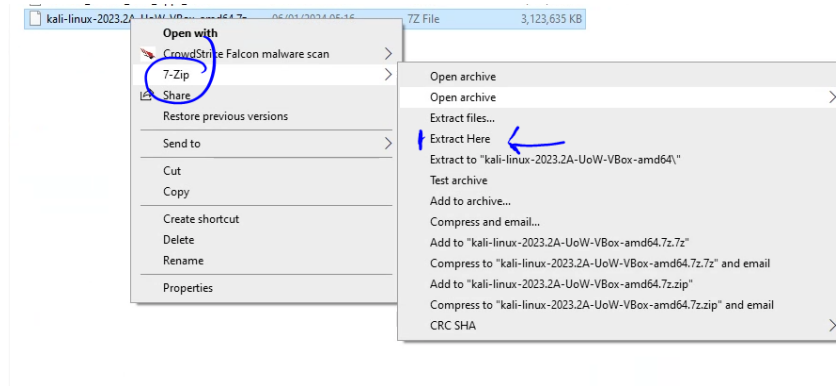


Figure 2: Extract the Kali 7zip file

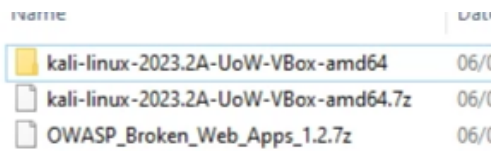


Figure 3: Kali Linux VM Folder after extraction

- (5) You will end up with a folder with the name as the file as shown in Fig.3
- (6) Now to do the same for the OWASP machine. Right Click on **OWASP_Broken_Web_Apps_1.2.7z** and select **7-Zip**, then select **Extract Here** and you will end up with the OWASP folder as well as shown in fig.4

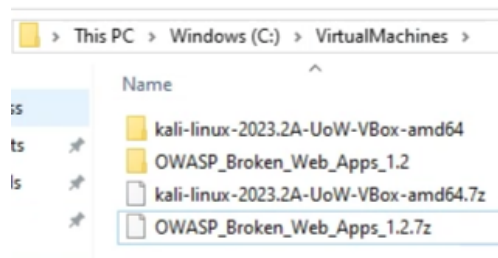


Figure 4: OWASP and Kali VMs Folders

- (7) Now you can move to section 3 to add the boxes to VirtualBox.

3 Setting up VirtualBox

- **If you are using a lab machine** It is already pre-installed on your lab machine.
- **If you are using your own machine** you should have installed it and set it up. See Section 1.1

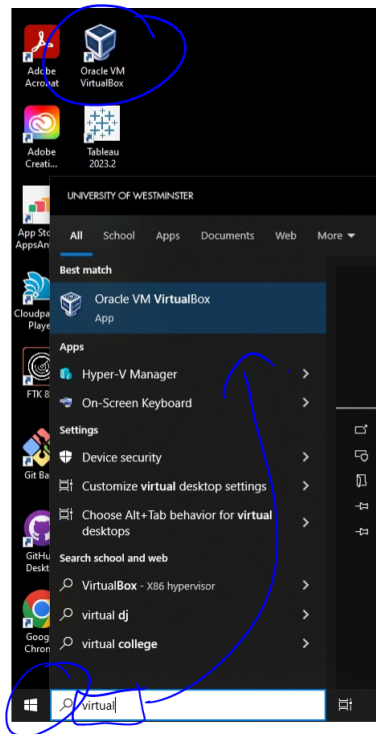


Figure 5: Start VirtualBox

- (1) You can now start the Oracle VirtualBox software by either double clicking on the Oracle VM Virtual Box shortcut on the Desktop or by clicking on the windows button and typing virtual box as shown in Fig.5

- (2) Once VirtualBox is running, Click on Add and
- (3) Browse to the virtual machines folder as shown in Fig.6

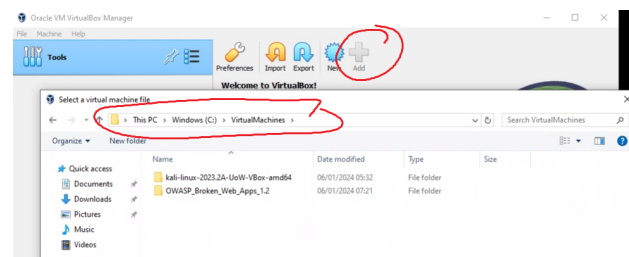


Figure 6: Locate VirtualMachines Fodler

- (4) Now let's start by **Adding** Kali Linux machine.
- (5) We browse to the Kali Linux folder and select "kali-linux-2023.2-virtualbox-amd64.vbox" and click open as shown in Fig.7
- (6) Now you have the Kali Linux machine added as shown in Fig.8

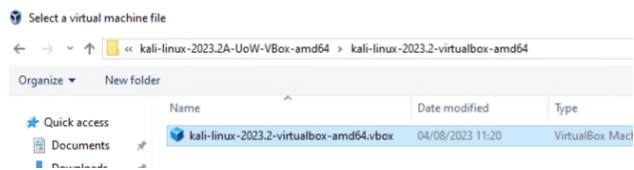


Figure 7: Add Kali Machine

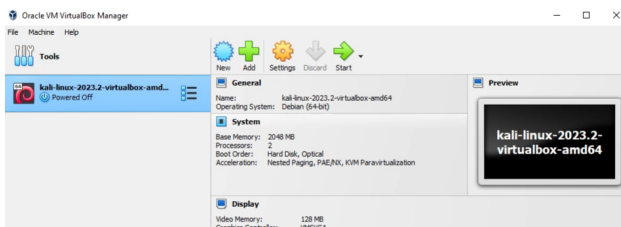


Figure 8: Kali VM added

(7) Now We do the same to add the OWASP VM.

(8) Click on Add

(9) Browse to the OWASP VM folder, select it and it and press **Open** as shown in Fig.9

(10) Now we have both machines setup in Virtual box as shown in Fig.10.

(11) We can now move to setting up the network settings for the lab environment in Section 4

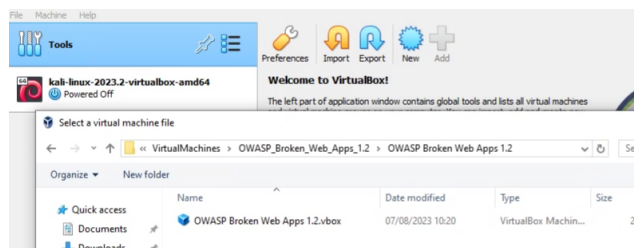


Figure 9: Add OWASP VM

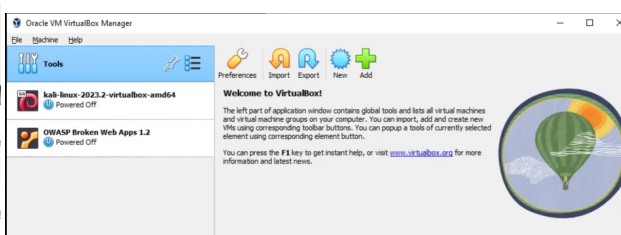


Figure 10: VMs setup on VirtualBox completed

4 Network Setup

- While working in the lab environment, you will need to ensure that it is isolated from the Internet
- One way of doing this is to create a private network
- All devices will join this network. isolated from the Internet.
- Sometimes, you need Kali to be connected to the Internet to download some scripts and tools or to conduct open Source Intelligence.
- However, it is essential that vulnerable machine VM stays on the private network.
- In this section we will look at:
 - We first need to create our private Virtual network environment as in section 4.1.
 - Once we have our private Virtual network environment, we can switch from Internet connected VMs to Private VMs.
 - * To connect a VM to the Internet, or to the Virtual Private network environment called in VirtualBox "**Host-Only Network**", see section 4.2

4.1 Private Virtual lab network environment

- Let us first both you will need to first create a Host-Only Network environment and set it up using the the following steps (shown in Fig.11):

Step.1- Click on the **three dots** next to tools.

- If you cannot see the three dots, make sure you click on tools

Step.2- Select **Network**

- If you find that you have a network already, you do not need to create it again. Just make sure the network IPv4 address is **192.168.56.1** as in step 3.

Step.3- Click on **Create**.

- If not, you can change it manually as per the setting shown in this step

Step.4- Click on **DHCP Server** and tick "Enable Server"

Step.5- Change the settings and make sure they are the same as below and click apply

- Server Address: 192.168.56.99
- Lower Address Bound: 192.168.56.100
- Upper Address Bound: 192.168.56.254
 - * This limit the IP addresses our VMs can possibly be allocated to each an IP address between 192.168.56.100 to 192.168.56.254

- Now we have our Host-Only Network network set, we can switch between **Host-Only** or Internet connectivity via **NAT** connection.

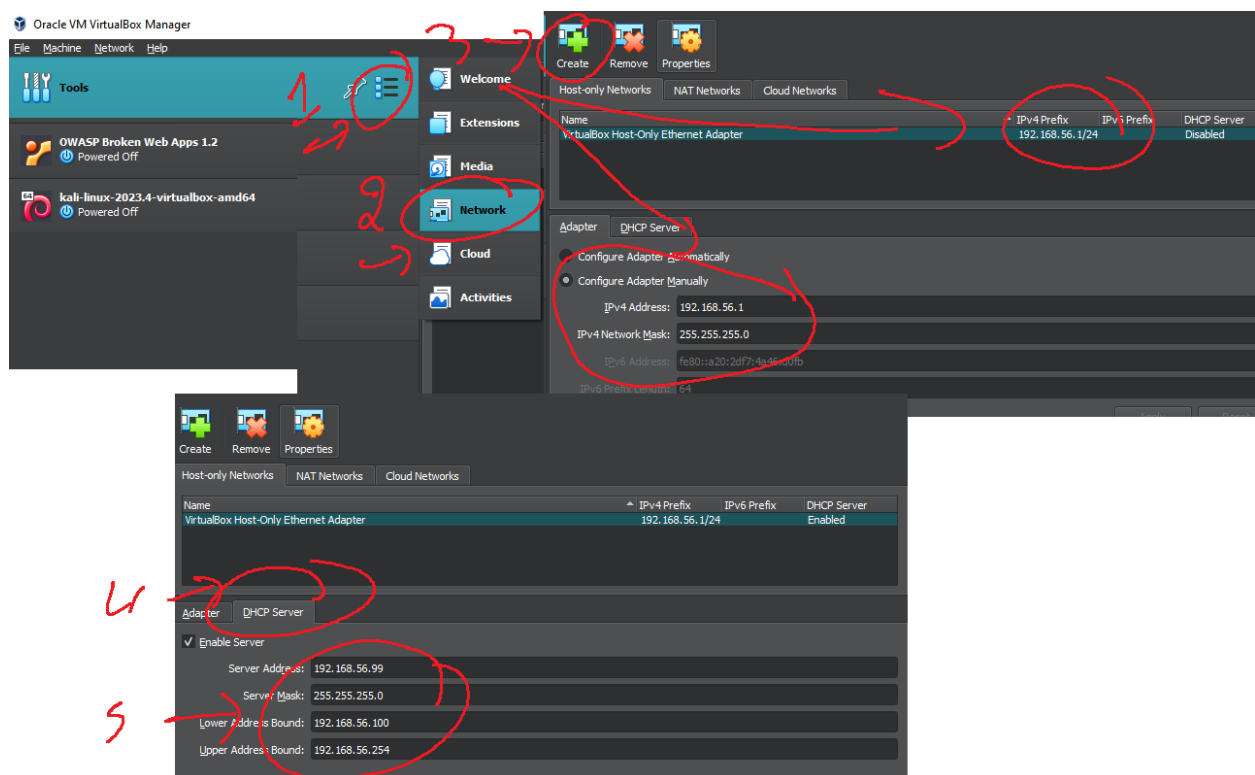


Figure 11: Host Only, DHCP network

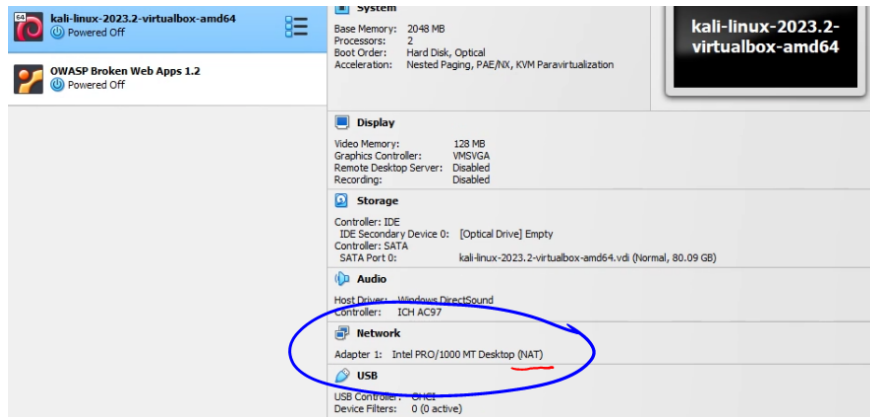


Figure 12: Kali Linux Default network Settings

4.2 Change VM Network Connection

- You can see which network the VM is connected to, simply by selecting it. For example, Kali by default is connected to the Internet via the **NAT** connection as shown in Fig.12
- To change the settings and switch to Host-Only (or the other way around), you can either click on **Settings** or **right click on the VM and select Settings** (Fig.13)
- Choose in the drop down menu of **Attached to** to the option **Host Only Adapter**. You should see the Network Name as **Virtual Box host-only adapter**. as shown in Fig 15

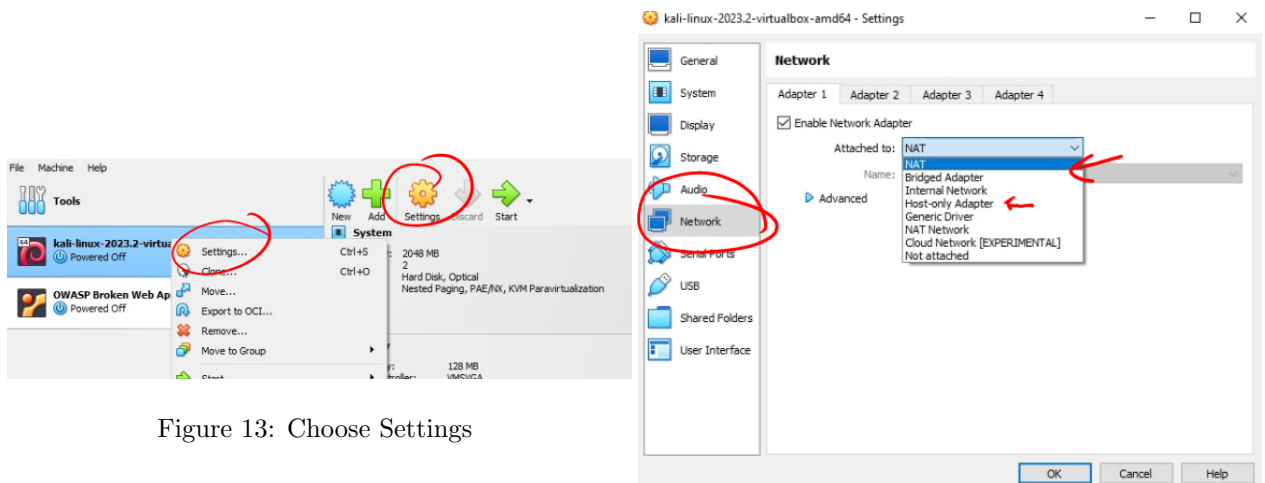


Figure 13: Choose Settings

Figure 14: Select Network and click on the network connectivity you need

- Choose in the drop down menu of **Attached to** to the option **NAT**. This is network Address Translation. This will give your virtual machine access to the network resources of your host machine. It will use the same IP address of your host machine to connect to the Internet. (Fig.14)
- **The steps shown in Figures 12, 13, 14, 15 need to be repeated for each VMs to ensure that all of them are on the same network or when you need a VM to be connected to the Internet.**

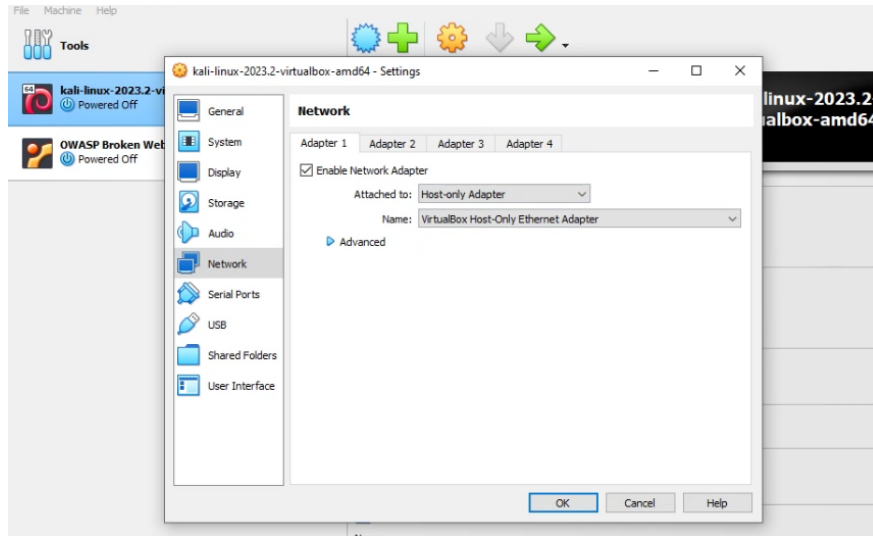


Figure 15: Host Only Adapter Settings

4.3 Start the VMs Test connectivity

VMs usernames and passwords

1. Kali Linux (Kali-linux-2022.3-UoW-virtualbox-amd64)
 - username: kali
 - password: kali
2. Owaspbwa linux (OWASP Broken Web Apps VM 1.2)
 - username: root
 - password: owaspbwa

- We can now start both VMs and see test connectivity.
- For each machine you want to start, either **double click on it**, or select it and click **Start**
- Enter the username and password for each VM
 - **Note:** On any Linux terminal, including the OWASP VM: When you enter a password, you will not see it on the terminal. The idea is to hide the length of the password if anyone is standing behind you and looking.
- To check your ip address, in any of those virtual machine devices you open a terminal and type:
 - **ifconfig**
- In Windows you can open a command line Interface by clicking on start and typing cmd.
 - **ipconfig**
- For example: on Kali Linux VM, we can check our IP as shown in the steps below (shown in Fig.16):

Step.1- Click on **the terminal Emulator Shortcut** to start the Terminal

Step.2- To check your IP address type **ifconfig**

- What does the output means:
 - * eth0 is the virtual network interface, the interface that connected our Virtual machine to the host (my physical machine): In my case, it is 192.168.56.100 - This means that my machine is connected to the Virtual private network we created earlier on in section 4.1
 - * lo is the localhost interface (also called loopback). the localhost IP 127.0.0.1 is an IP used for testing services on the machine. if you connect to it, it loops back to your machine. We don't need it now, but it will appear when you type **ifconfig**

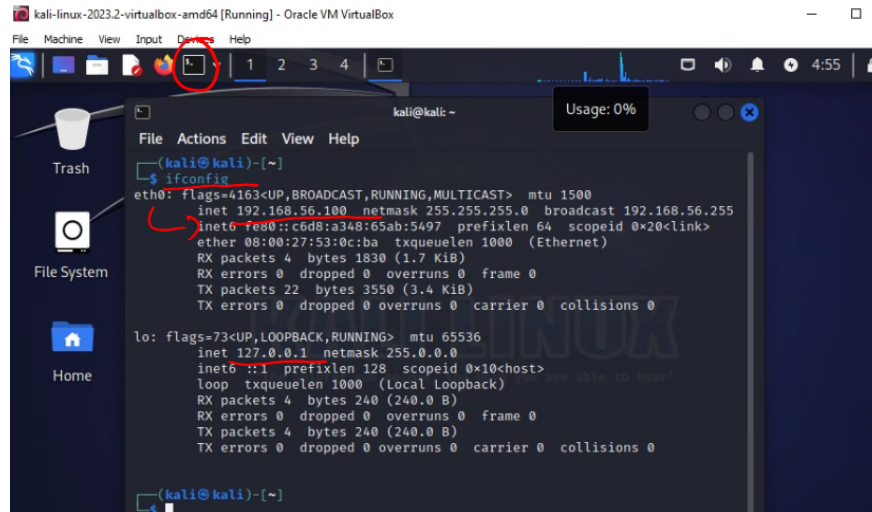


Figure 16: Check IP address on Kali

- All the steps should be done on the OWASP VM and on any other VM you can possibly connect.

Change IP without restarting

- You do not need to restart the machine to change the network settings.
- What you need to do is the following:

Step.1- Click on **Machine** as shown in Fig.17

Step.2- Select **Settings** as shown in Fig.17

Step.3- Choose "Change adapter Settings" to "Host-only Adapter, or to "NAT" as shown in Fig.17

Step.4- Click **ok**

Step.5- open Terminal and type

sudo ifconfig eth0 down

- * This will disable the eth0 interface as shown in Fig.18

sudo ifconfig eth0 up

- * This will enable the eth0 interface again with a new IP as shown in Fig.18

- Check IP address again:

ifconfig

- IP changed to NAT IP as shown in Fig.18

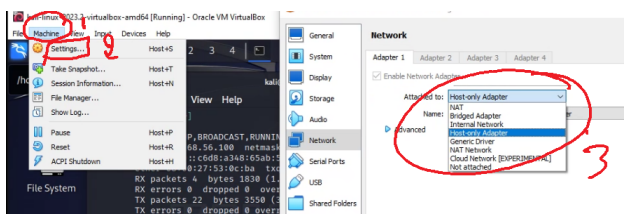


Figure 17: Change IP without restarting

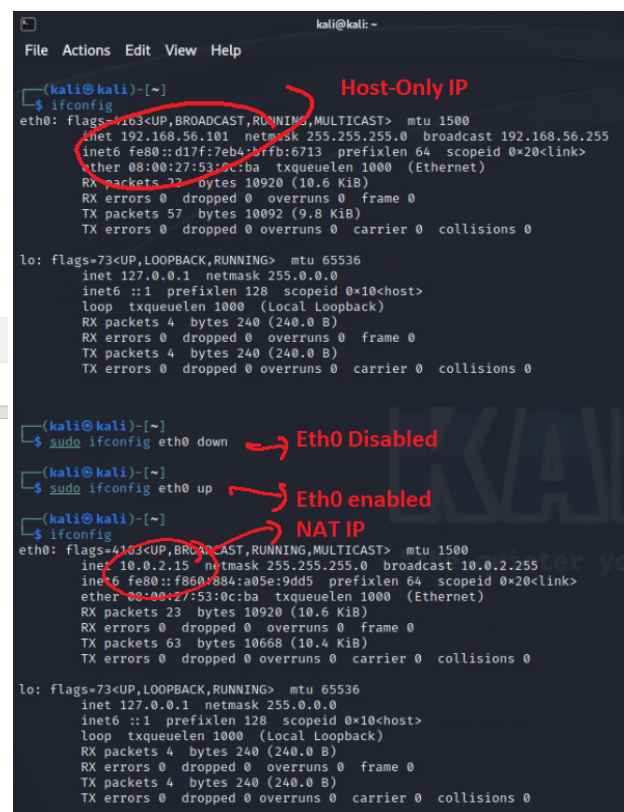


Figure 18: Check IP address on Kali

4.4 Test connectivity

- To test connection between devices, they **MUST** be on the same network.
- We can test connectivity between machines on the terminal by using **ping** command both on Windows or Linux.
- **Note:** You will need to check your IP addresses before you do this step. Your VMs IP addresses might be different then the one in the examples below. My IP addresses are shown in Fig.19

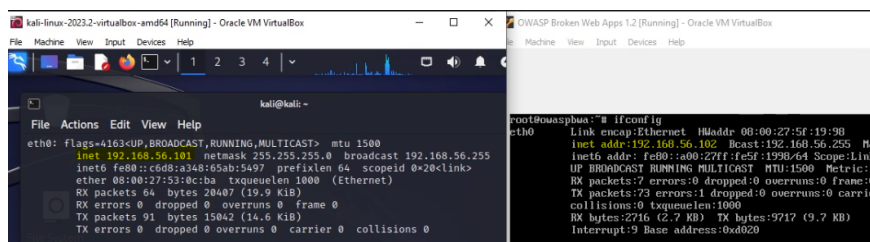


Figure 19:

- To check connectivity between devices:
 - To ping the vulnerable machine from kali:

ping 192.168.56.102
 - To ping kali machine from the vulnerable machine

ping 192.168.56.101

- when we selected host only network we created a private network between our VMs that is completely isolated from the external world.
- Your host machine also have a virtual Box interface that connects your device to the private network.
 - To ping your host machine (physical) machine from the kali machine:

ping 192.168.56.1

- In Fig.20, you can see my lab environment setup based on the tests we did now.

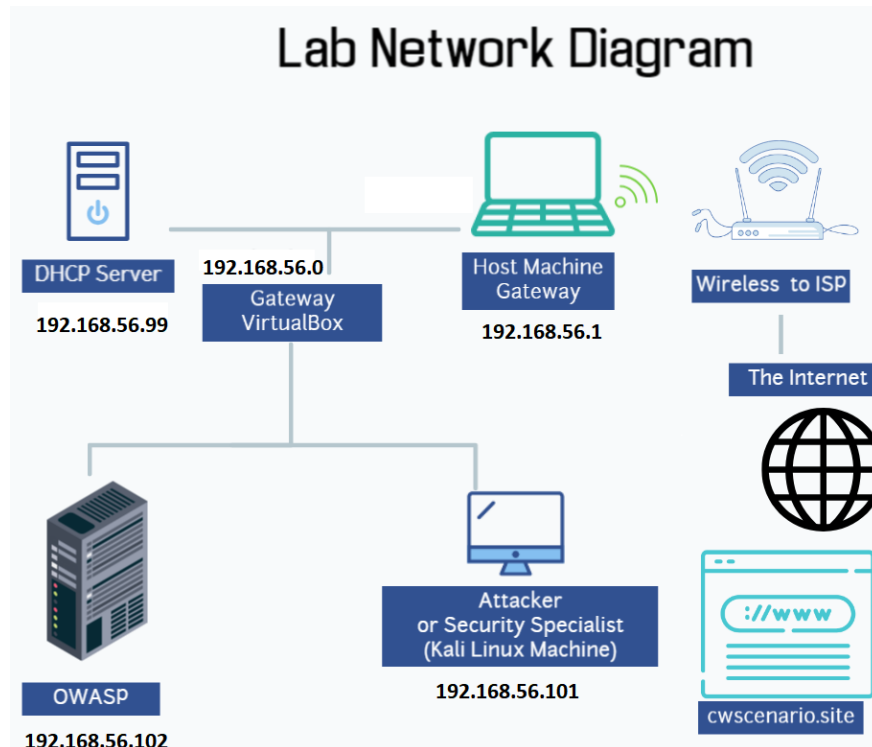


Figure 20: Check IP address on Kali

Instructions

Setup for the Vulnerable Juice shop server

- The penetration test for your assessment should be conducted on the Juice Shop website (locally installed)
 - Instructions on how to install the Juice Shop website site on Kali VM is provided below.
 - No further information including passwords will be provided.
-
- This assignment only needs Kali Linux VM and Burpsuite.
 - Make sure you have Kali Linux VM running and the network interface is set on NAT.

- Before we start installing, lets update Kali repository

sudo apt-get update

- Install Docker. We need it to use The vulnerable machine.

sudo apt-get install docker-compose docker.io

- go to your home folder.

cd ~

- Download the vulnerable Juice shop website server **Juice Shop**

sudo docker pull bkimminich/juice-shop

- Optional: You can add a local domain name to hosts file. This way, you can access your server by pointing the browser to <http://juiceshop.local> or anything else you want

sudo sed -i '\$a127.0.0.1 juiceshop.local' /etc/hosts

- Run it

sudo docker run --rm -p 3000:3000 bkimminich/juice-shop

- Now the Juice Shop server is running, open your browser and point it to <http://juiceshop.local:3000> or 127.0.0.1:3000

Extra machine to test your skills DVWS

- The penetration test should be conducted on the Juice Shop website (locally installed)
- Instructions on how to install the Juice Shop website site on Kali VM is provided below.
- No further information including passwords will be provided.

- This assignment only needs Kali Linux VM and Burpsuite.
- Make sure you have Kali Linux VM running and the network interface is set on NAT.
- go to your home folder.

cd ~

- Download the vulnerable API server **DVWS**

git clone https://github.com/snoopysecurity/dvws-node.git

- Optional: You can add a local domain name to hosts file. This way, you can access your server by pointing the browser to <http://dvws.local>

sudo sed -i '\$a127.0.0.1 dvws.local' /etc/hosts

- Browse to DVWS folder

cd dvws-node

- Run it

sudo docker-compose up

- Now the DVWS server is running, open your browser and point it to <http://dvws.local>

5 Problems that you can possibly face

Problems you can possibly have, **only on your personal machine:**

1. Sometimes, the machines do not start and give an error. It will tell you what the problem is: Usually, it is:
 - Either no space for the machines to start. You should free up some space for the machines.
 - or VM virtualisation should be enabled from BIOS. Check this [link from Microsoft](#) on how to do this. If you need help, please ask your lab tutor.
2. If you cannot see Host-Only adapter, it means you did not create the network. On Vbox, Make sure you created the virtual Host only network interface first.
3. On some machines, you need to change the USB setting for the VM to USB 1.1 as shown in fig below. This applies for any machines that throw the USB error.
4. if the network do not change ONLY from NAT to Host-Only, you can type **sudo dhclient**

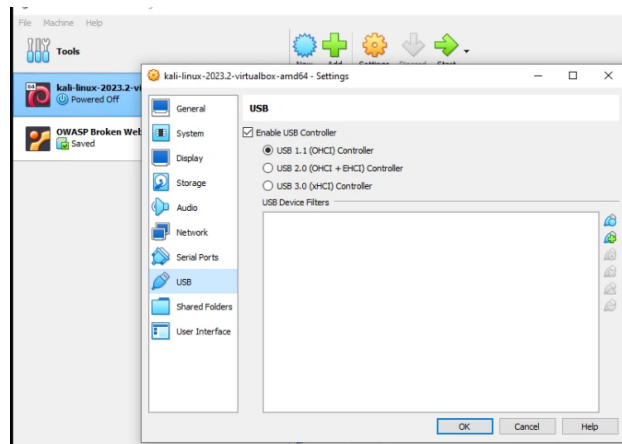


Figure 21: USB error